Sustainable Transportation Coordinator

Workshop #2

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Executive summary

During the first workshop, an analysis of the transportation and parking system of the Faculty of Engineering of the Distrital Francisco José de Caldas University was proposed, where some main problems were evident, such as the short time between changes, the lack of interest and real solution on the part of the university, the economic issue, the lack of spaces and parking for students. It is sought through a real analysis from a holistic perspective of what our system is, along with sensitivity and some other factors, some survey-type mechanisms were carried out to see the opinion of the students, where some opinions were asked about the means of transportation used and what was their acceptance with some possible solutions. Rejection was observed to the shared bicycle proposal, while the idea of carpooling was quite well received, this being an important and essential factor for the future of what will be the solution and analysis of this problem.

Requirements Analysis

Student Profile Verification. The system shall verify users through their institutional email to ensure a trusted community.

Publication of routes (drivers). Students with a car or motorcycle will be able to post their route, departure time, and available seats.

Passenger Search. Students will be able to search for available trips by filtering by campus destination (Calle 40, Calle 34 or ECCI) and arrival time.

Chat/notification in real time. The application will have a way of communication between the driver and the passenger to coordinate the meeting point.

Scalability. The backend must be able to handle the increase in trips and simultaneous users during peak hours such as between 8:00 a.m. and 8:00 p.m. m. and 2:00 p.m. m or moments such as rain or factors that make students need carpooling.

Data privacy. The system must encrypt sensitive user data (phone numbers, license plate numbers) and comply with data protection.

Reliability. The database should implement connection pooling to prevent system failures during high demand logging periods.

Complexity and Risk Mitigation

Sensitivity Management (Security & Trust)

To address student safety, the system implements a Mandatory Institutional Verification. Only users with a verified email can access the matching module. A Reputation Scoring System is integrated to ensure accountability and build trust within the university community.

Complexity Handling (Traffic & Manual Updates)

Given that traffic in the Calle 40 area is unpredictable, the design includes a Manual Status Update Module.

System Resilience (Failure Modes & Communication)

To mitigate risks during network outages, the system implements a Pre-trip Information Protocol. Once a match is confirmed, the application displays a summary including the driver's name and vehicle plate. Users are encouraged to record this information manually. On the server side, a Transaction Log within the Persistence Layer ensures that active trip data can be recovered and synchronized in case of a temporary server restart.

System Architecture

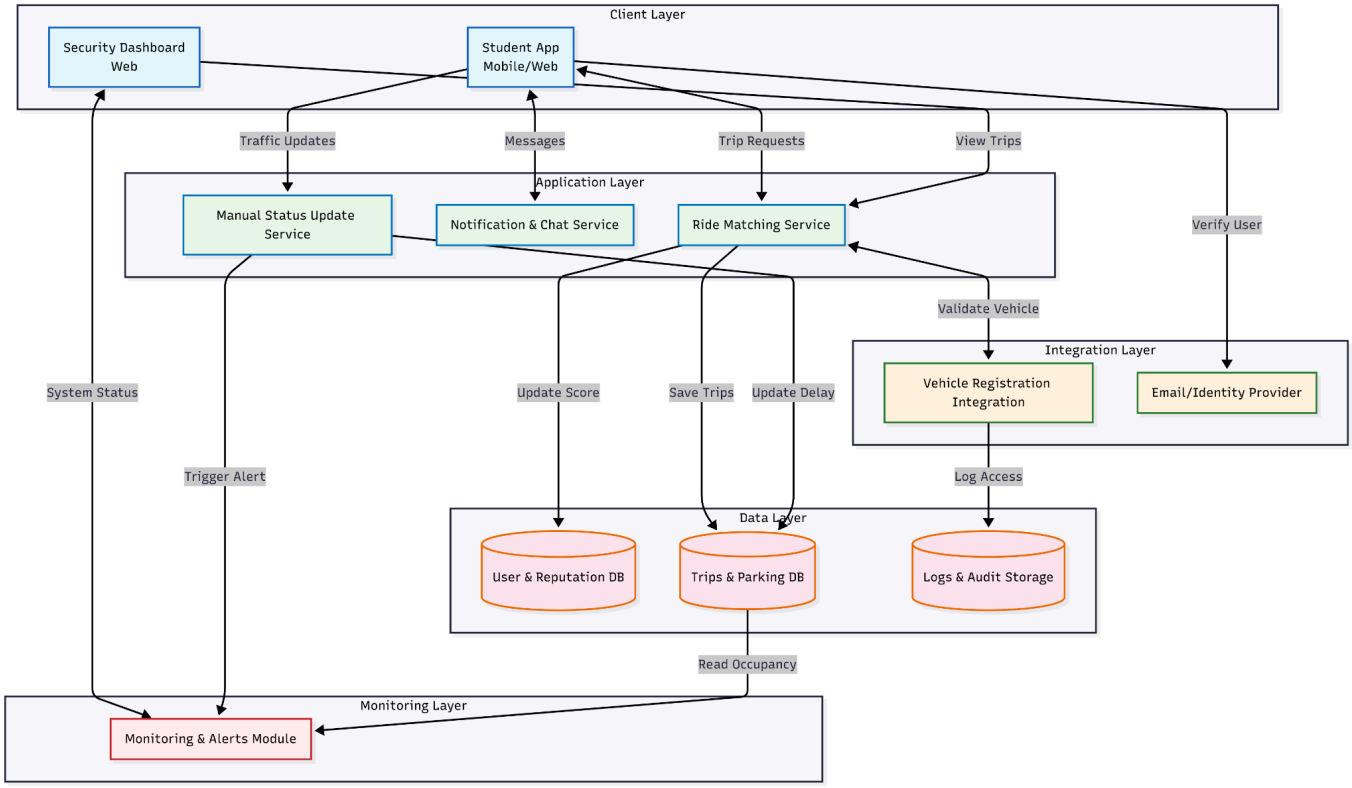

Figure 1. System architecture diagram illustrating the interaction between client interfaces, application services, integration modules, and data layers

The proposed architecture is a layered system designed to solve the communication and the bottlenecks identified in the first workshop. The Client Layer provides tailored interfaces: a mobile app for students to replace informal WhatsApp groups, and a web dashboard for security staff to digitize access control

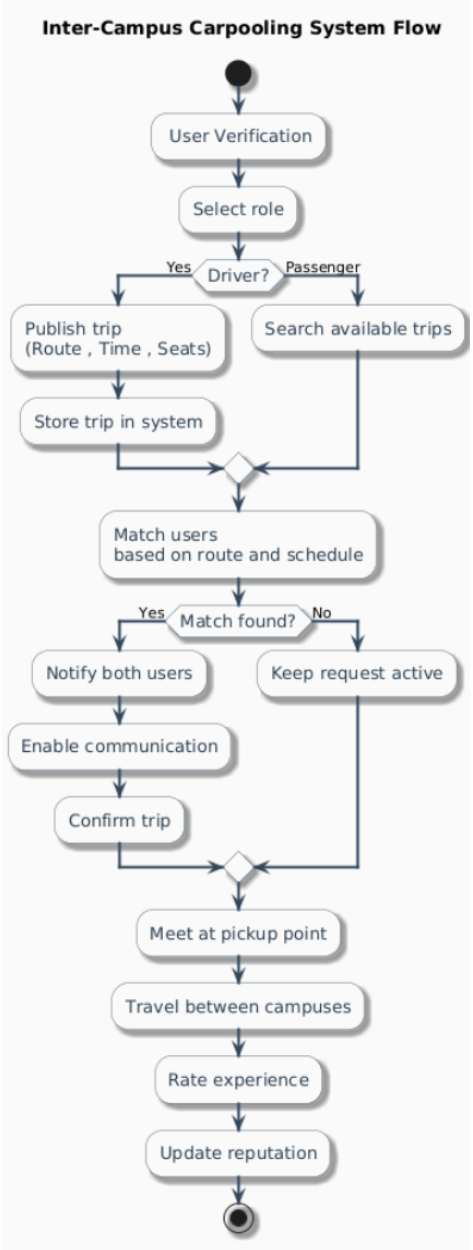

Figure 2 System flowchart

The flowchart represents the operation of the ride-sharing system, starting with user verification and role selection, followed by the trip search and connection process between drivers and users, then coordination and finally the system records the trip and updates the user's reputation for future interactions.

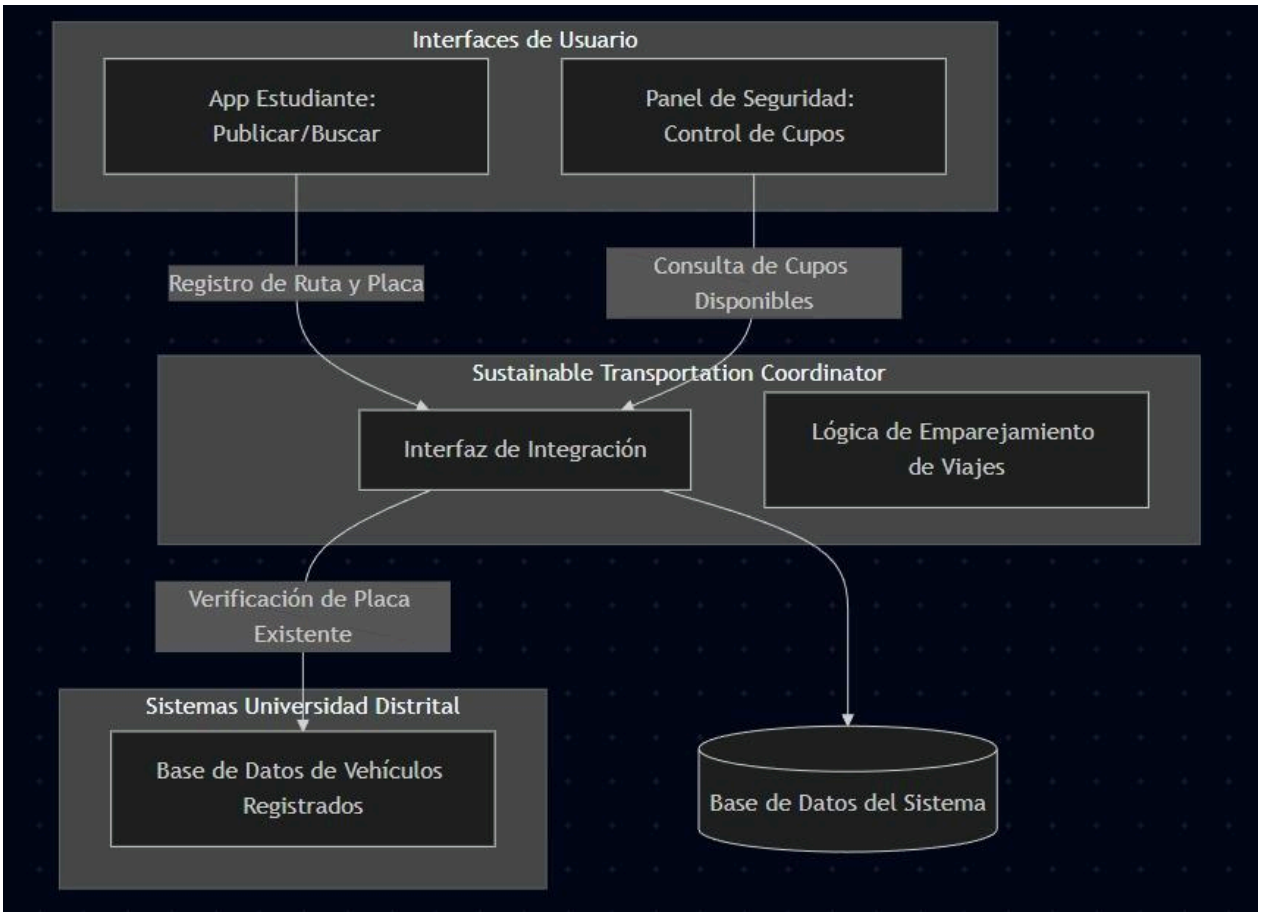

Figure 3. Integration and data flow architecture of the system.

This figure illustrates how the client interfaces (Student App and Security Dashboard) communicate with the system's core. The integration interface routes user requests to the ride-matching logic while securely verifying vehicle information against the university's external database and storing transaction records locally to ensure data persistence.

Implements Strategy

Implementation Strategy

1. General Approach

The system implementation strategy is based on the development and analysis of the operation of a shared transportation platform (carpooling) that helps students coordinate trips between campuses, saving time and money.

This seeks to replace the methods that are used (such as the use of WhatsApp or Facebook groups) with a better and more efficient solution that improves communication and the possibility of achieving this mechanism.

2. Implementation Phases

The system will be carried out in the following phases:

Phase 1: Design Definition of what the program will really be like that will help with student communication.

Phase 2: Development of the proposed ideas and linking them to the needs of the students.

Phase 3: Test

Simulations with several users and real system functionality.

3. Integration Strategy

The system should help by:

Real-time communication between users and Immediate update of available trips.

4. Considerations

Limited resources:

A solution is sought for multiple problems without a large use of resources.

5. Relationship with System Analysis (Workshop 1)

This strategy is based on the findings identified in the analysis of the workshop:

The lack of coordinated communication solved through a platform

The use of informal means replaced by a better structured system.

The need for transportation between locations that seeks to be solved through carpooling

Conclusion

The carpooling system provides a solution to the transportation problems identified in the initial analysis and with slight changes according to the information obtained and analyzed in workshop #1. By avoiding the old ways of coordinating and replacing it with our system with which communication is improved, it reduces travel uncertainty and makes movement between locations more effective.

The design focuses on some important and key principles of systems engineering, such as scalability, reliability, security, synergy, which seeks to solve this problem at our university. In general, this work demonstrates how a system with good design and analysis can improve a problem such as student transportation in our faculty.